

```
!pip install pandas
!pip install matplotlib
!pip install seaborn
!pip install numpy
```

Defaulting to user installation because normal site-packages is not writeable

Requirement already satisfied: pandas in c:\programdata\anaconda3\lib\site-packages (2.3.3)

Requirement already satisfied: numpy>=1.26.0 in c:\programdata\anaconda3\lib\site-packages (from pandas) (2.3.5)

Requirement already satisfied: python-dateutil>=2.8.2 in c:\programdata\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in c:\programdata\anaconda3\lib\site-packages (from pandas) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in c:\programdata\anaconda3\lib\site-packages (from pandas) (2025.2)

Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

Defaulting to user installation because normal site-packages is not writeable

Requirement already satisfied: matplotlib in c:\programdata\anaconda3\lib\site-packages (3.10.6)

Requirement already satisfied: contourpy>=1.0.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (1.3.3)

Requirement already satisfied: cycler>=0.10 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (1.4.9)

Requirement already satisfied: numpy>=1.23 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (2.3.5)

Requirement already satisfied: packaging>=20.0 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (25.0)

Requirement already satisfied: pillow>=8 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (12.0.0)

Requirement already satisfied: pyparsing>=2.3.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (2.9.0.post0)

Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)

Defaulting to user installation because normal site-packages is not writeable

Requirement already satisfied: seaborn in c:\programdata\anaconda3\lib\site-packages (0.13.2)

Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\programdata\anaconda3\lib\site-packages (from seaborn) (2.3.5)

Requirement already satisfied: pandas>=1.2 in c:\programdata\

```

anaconda3\lib\site-packages (from seaborn) (2.3.3)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\
programdata\anaconda3\lib\site-packages (from seaborn) (3.10.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\programdata\
anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn)
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Requirement already satisfied: numpy in c:\programdata\anaconda3\lib\
site-packages (2.3.5)

```

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

```

```

df=pd.read_csv('Customer Churn.csv')
df.head()

```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure
PhoneService	\					
0	7590-VHVEG	Female	0	Yes	No	1
No						

1	5575-GNVDE	Male	0	No	No	34
Yes						
2	3668-QPYBK	Male	0	No	No	2
Yes						
3	7795-CF0CW	Male	0	No	No	45
No						
4	9237-HQITU	Female	0	No	No	2
Yes						

MultipleLines		InternetService	OnlineSecurity	...
DeviceProtection \				
0	No phone service	DSL	No	...
No				
1	No	DSL	Yes	...
Yes				
2	No	DSL	Yes	...
No				
3	No phone service	DSL	Yes	...
Yes				
4	No	Fiber optic	No	...
No				

TechSupport		StreamingTV	StreamingMovies	Contract
PaperlessBilling \				
0	No	No	No	Month-to-month
Yes				
1	No	No	No	One year
No				
2	No	No	No	Month-to-month
Yes				
3	Yes	No	No	One year
No				
4	No	No	No	Month-to-month
Yes				

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

[5 rows x 21 columns]

df.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 7043 entries, 0 to 7042

Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
---	--------	----------------	-------

```

---  -----
0   customerID      7043 non-null  object
1   gender          7043 non-null  object
2   SeniorCitizen   7043 non-null  int64
3   Partner         7043 non-null  object
4   Dependents      7043 non-null  object
5   tenure          7043 non-null  int64
6   PhoneService    7043 non-null  object
7   MultipleLines   7043 non-null  object
8   InternetService 7043 non-null  object
9   OnlineSecurity  7043 non-null  object
10  OnlineBackup    7043 non-null  object
11  DeviceProtection 7043 non-null  object
12  TechSupport     7043 non-null  object
13  StreamingTV     7043 non-null  object
14  StreamingMovies 7043 non-null  object
15  Contract        7043 non-null  object
16  PaperlessBilling 7043 non-null  object
17  PaymentMethod   7043 non-null  object
18  MonthlyCharges  7043 non-null  float64
19  TotalCharges    7043 non-null  object
20  Churn           7043 non-null  object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

```

Replacing blanks with 0 as tenure is 0 and no total charges are recorded

```

df["TotalCharges"] = df["TotalCharges"].replace(" ", "0")
df["TotalCharges"] = df["TotalCharges"].astype("float")

df.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   customerID          7043 non-null  object
1   gender              7043 non-null  object
2   SeniorCitizen        7043 non-null  int64
3   Partner             7043 non-null  object
4   Dependents          7043 non-null  object
5   tenure              7043 non-null  int64
6   PhoneService         7043 non-null  object
7   MultipleLines        7043 non-null  object
8   InternetService      7043 non-null  object

```

```

9   OnlineSecurity      7043 non-null object
10  OnlineBackup        7043 non-null object
11  DeviceProtection    7043 non-null object
12  TechSupport         7043 non-null object
13  StreamingTV         7043 non-null object
14  StreamingMovies     7043 non-null object
15  Contract            7043 non-null object
16  PaperlessBilling    7043 non-null object
17  PaymentMethod       7043 non-null object
18  MonthlyCharges      7043 non-null float64
19  TotalCharges        7043 non-null float64
20  Churn               7043 non-null object

```

```
dtypes: float64(2), int64(2), object(17)
```

```
memory usage: 1.1+ MB
```

```
df.isnull().sum().sum()
```

```
np.int64(0)
```

```
df.describe()
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2266.794470
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

```
df["customerID"].duplicated().sum()
```

```
np.int64(0)
```

#converted 0 and 1 values of Senior Citizens to yes/no to make it easier to understand

```

def conv(value):
    if value == 1:
        return "yes"
    else:
        return "no"

```

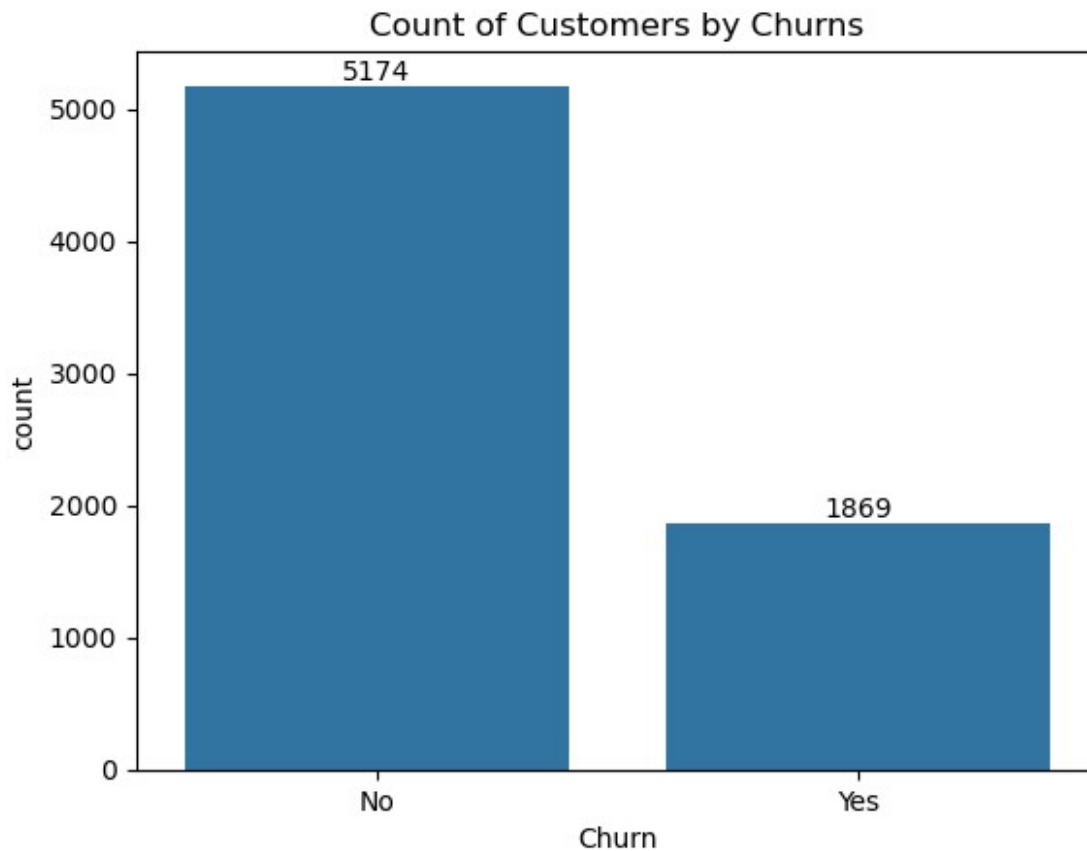
```
df["SeniorCitizen"] = df["SeniorCitizen"].apply(conv)
```

```

ax = sns.countplot(x= 'Churn', data = df)
ax.bar_label(ax.containers[0])
plt.title("Count of Customers by Churns")
plt.show

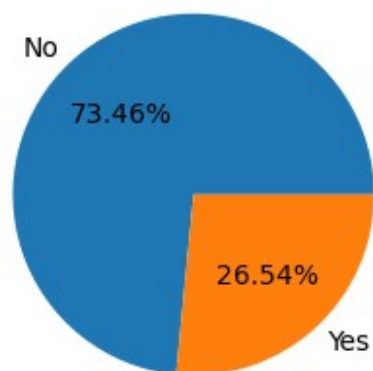
```

```
<function matplotlib.pyplot.show(close=None, block=None)>
```



```
plt.figure(figsize=(3,4))
gb = df.groupby("Churn").agg({'Churn' : "count"})
plt.pie(gb['Churn'], labels= gb.index, autopct = "%1.2f%%")
plt.title("Percentage of Churned Customer")
plt.show()
```

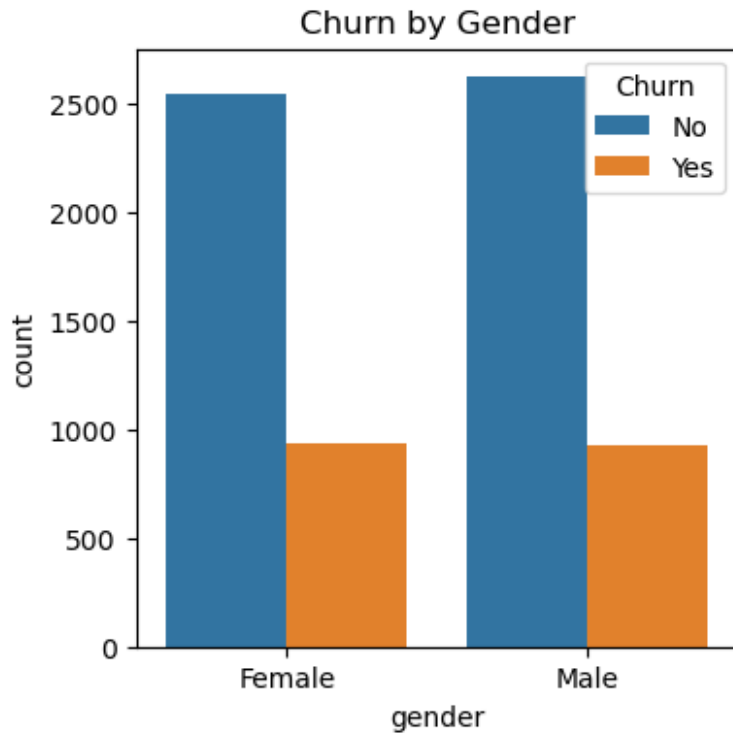
Percentage of Churned Customer



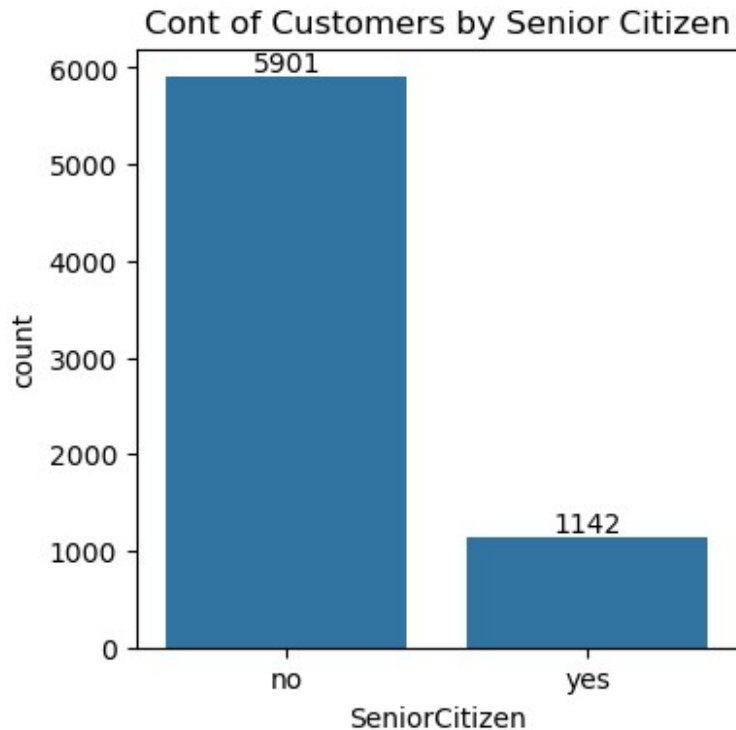
#From the given pie chart we can conclude that 26.54% of our customers have churned out.

#Now let Explore the reason behind it

```
plt.figure(figsize=(4,4))
sns.countplot(x="gender" , data = df, hue="Churn")
plt.title("Churn by Gender")
plt.show()
```



```
plt.figure(figsize=(4,4))
ax = sns.countplot(x="SeniorCitizen" , data = df)
ax.bar_label(ax.containers[0])
plt.title("Cont of Customers by Senior Citizen")
plt.show()
```

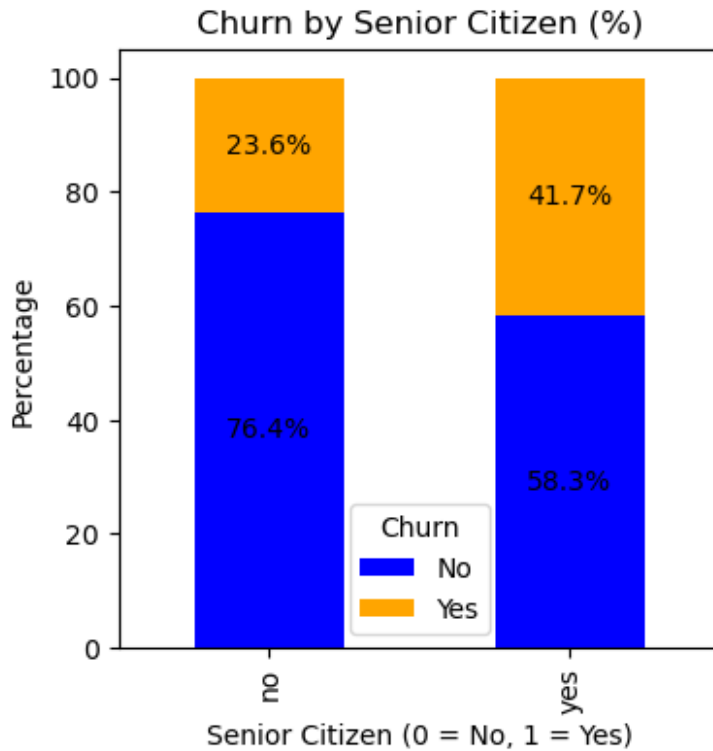


```
# Crosstab and convert to %
ct = pd.crosstab(df['SeniorCitizen'], df['Churn'], normalize='index')
* 100

# Plot
ax = ct.plot(kind='bar', stacked=True, color=['blue', 'orange'],
figsize=(4,4))

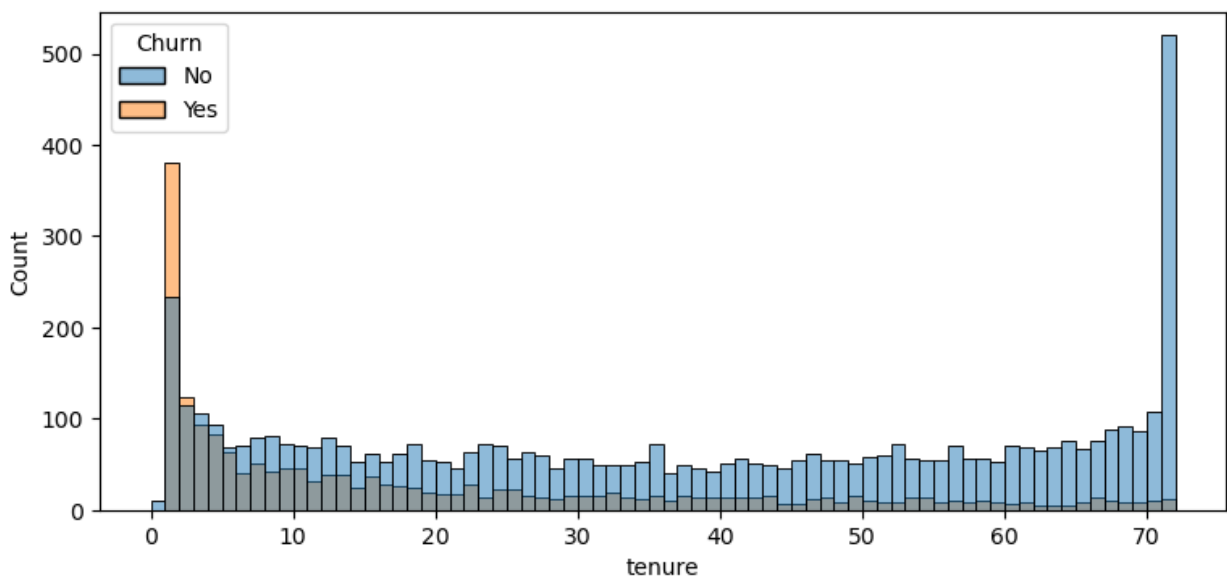
# Add % labels
for i in range(len(ct)):
    cum = 0
    for val in ct.iloc[i]:
        ax.text(i, cum + val/2, f"{val:.1f}%", ha='center',
va='center')
        cum += val

plt.title("Churn by Senior Citizen (%)")
plt.xlabel("Senior Citizen (0 = No, 1 = Yes)")
plt.ylabel("Percentage")
plt.legend(title="Churn")
plt.show()
```



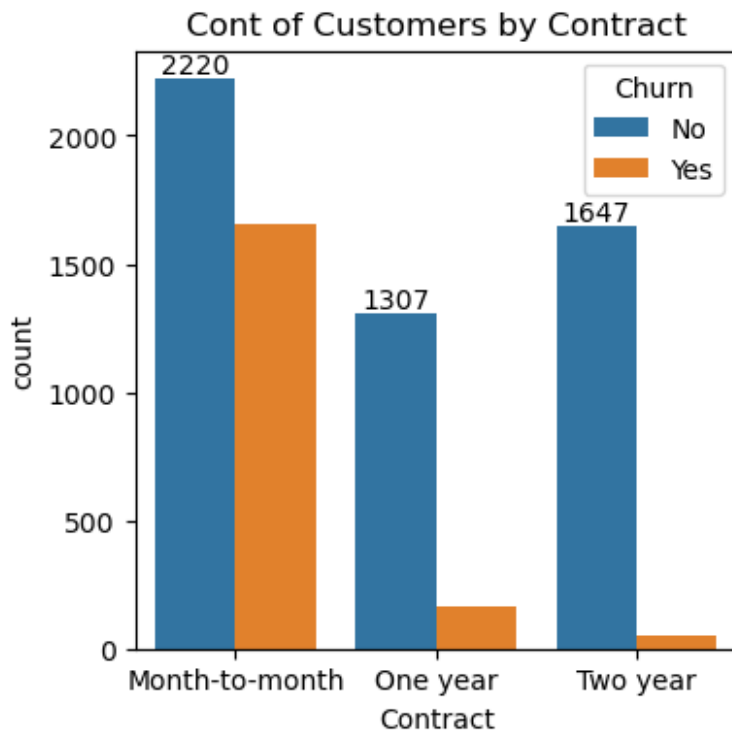
#comparative a greater percentage of people in senior citizen category have churned

```
plt.figure(figsize=(9,4))
sns.histplot(x= "tenure",data=df, bins=72, hue = "Churn")
plt.show()
```



#people who have used our services for a long time have stayed and people who have used our services 1 or 2 months have churned

```
plt.figure(figsize=(4,4))
ax = sns.countplot(x="Contract" , data = df, hue="Churn")
ax.bar_label(ax.containers[0])
plt.title("Cont of Customers by Contract")
plt.show()
```



#people who have month to month contract are likely to churn than from those who have 1 or 2 years of contract

```
df.columns.values
array(['customerID', 'gender', 'SeniorCitizen', 'Partner',
      'Dependents',
      'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
      'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
      'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',
      'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges',
      'TotalCharges', 'Churn'], dtype=object)

cols = ['PhoneService', 'MultipleLines', 'InternetService',
        'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
        'TechSupport', 'StreamingTV', 'StreamingMovies']

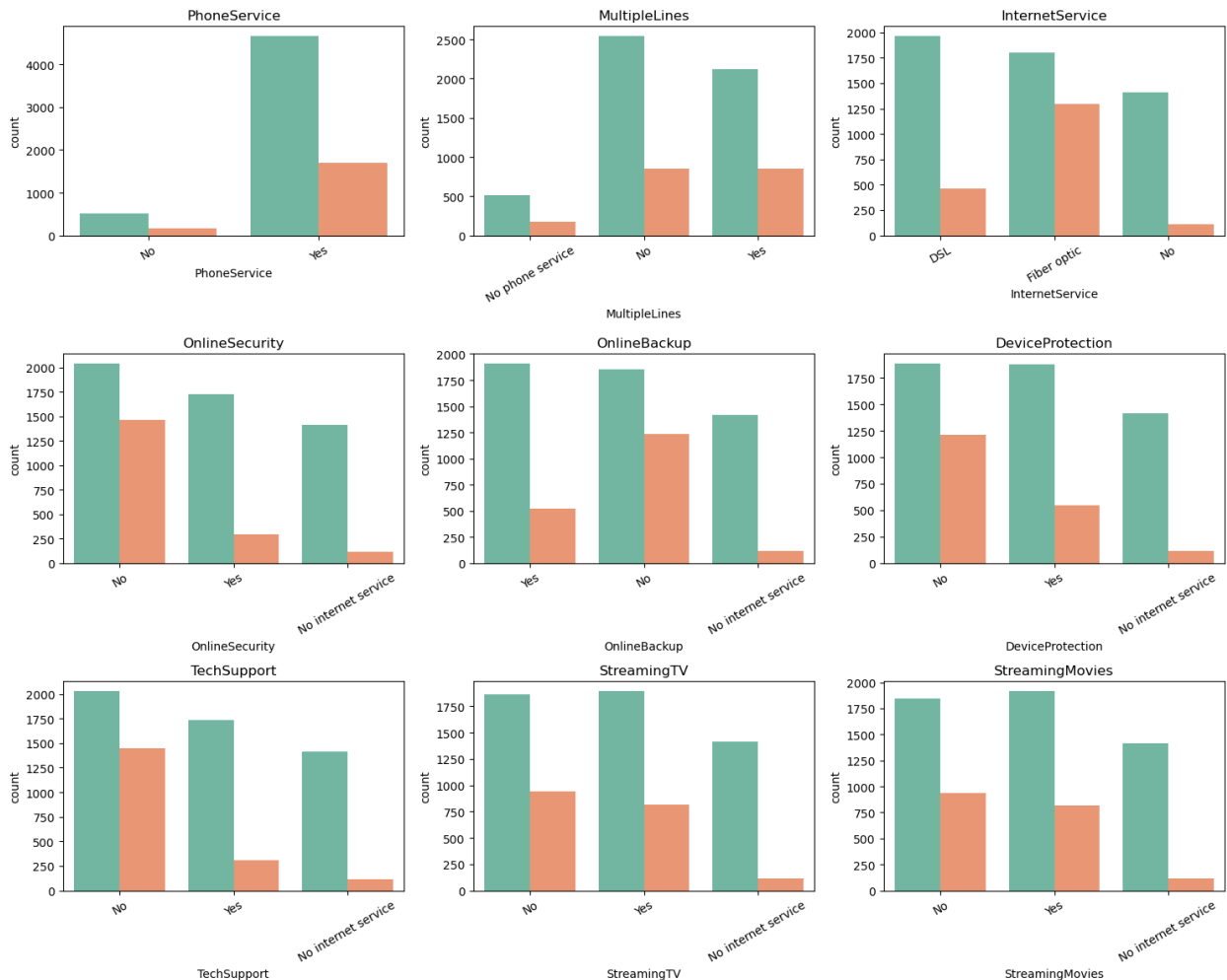
fig, axes = plt.subplots(3, 3, figsize=(15, 12))
axes = axes.flatten()

for i, col in enumerate(cols):
```

```
sns.countplot(data=df, x=col, hue=df["Churn"], ax=axes[i],
              palette='Set2', legend=False)
```

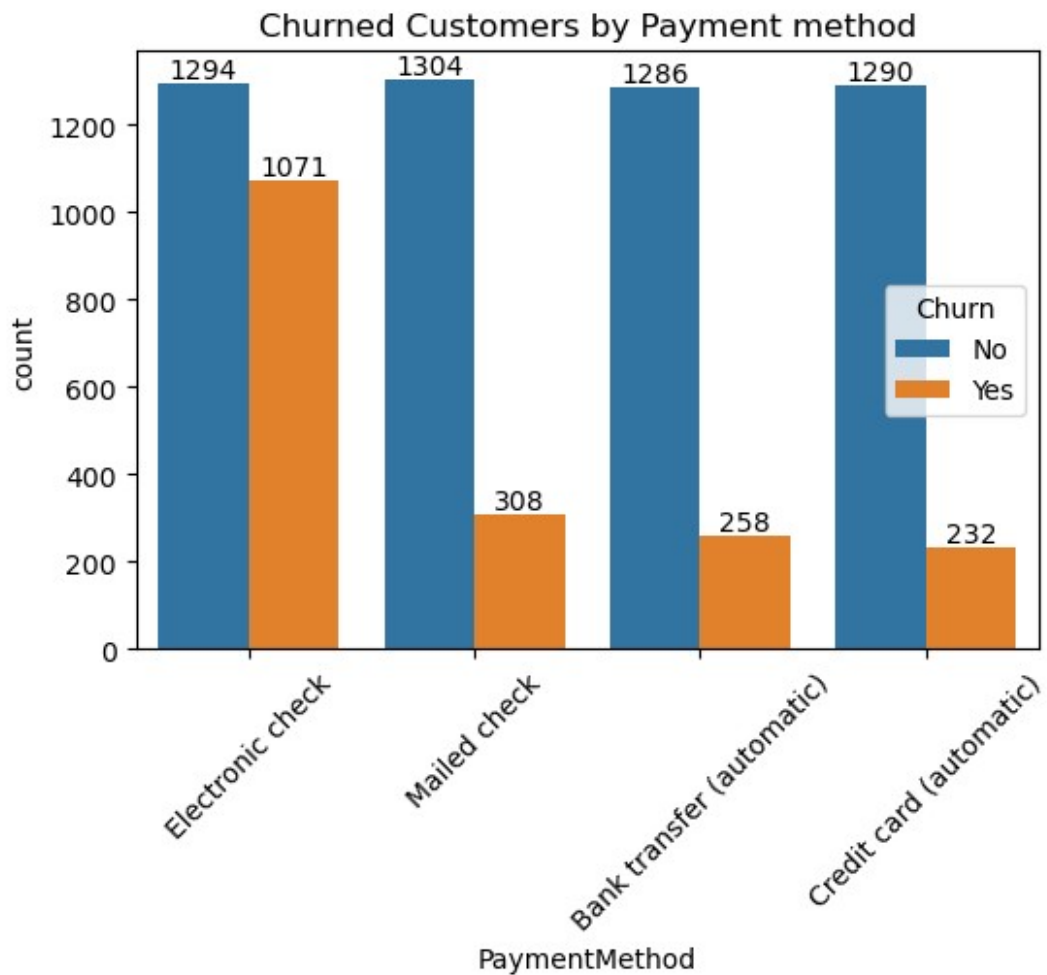
```
axes[i].set_title(col)
axes[i].tick_params(axis='x', rotation=30)
```

```
plt.tight_layout()
plt.show()
```



The data shows that while customers using basic telecom services like phone service and multiple lines form the majority, a noticeable portion of them still churn. A clear pattern emerges where customers who do not use add-on services such as online security, tech support, and device protection are more likely to leave, indicating the importance of these services in customer retention. Additionally, users with fiber optic internet exhibit relatively higher churn compared to DSL users, suggesting possible dissatisfaction despite better connectivity. Overall, the absence of value-added services and certain service types plays a key role in increasing customer churn.

```
plt.figure(figsize=(6,4))
ax = sns.countplot(x= "PaymentMethod", data = df, hue = "Churn")
ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])
plt.title("Churned Customers by Payment method")
plt.xticks(rotation = 45)
plt.show()
```



#Customer is likely to churn when he using electronics check as a payment method.